

MEACompass

Reliability-Aware Early Prediction of Neural Network Development from Microelectrode-Array Assays

Toward Functional Digital Twins for Neural Organ-on-Chip Screening

Team information

Author: Ziyad Azzaz — College of Artificial Intelligence, Arab Academy for Science, Technology & Maritime Transport (AASTMT), Alamein Campus, Egypt.

Team name: USER CONFIRMATION REQUIRED. No additional team member or affiliation is inferred.

Abstract

Late functional readouts slow developmental-neurotoxicity triage. MEACompass forecasts day-in-vitro 12 (DIV12) neural microelectrode-array outcomes from information available by DIV7, attaches a calibrated interval, and abstains on the least certain 30% of cases. Evaluation is chemical-disjoint: all doses, wells, and replicates of a chemical remain together across three repetitions of five outer folds. On five preregistered endpoints, the frozen model reduced mean absolute error by 14.2–39.0% relative to BT+, a strong baseline chosen by inner validation only; all chemical-bootstrap 95% confidence intervals for paired error differences excluded zero. Against the stricter post-audit BT++ comparator, the corresponding improvement was 12.3–42.9%, again significant on all five endpoints. Nested group CV+ achieved 91.0–92.4% empirical coverage for nominal 90% intervals. At 70% retained coverage, error fell materially for firing rate (39.29 to 31.87), active electrodes (20.42 to 16.75), and coordinated activity (48.61 to 34.81), while burst and network-spike reductions did not reach the registered 15% threshold. An initially positive permutation control triggered a registered stop; a bounded integrity audit found no feature leakage, showed a clean full-shuffle control, and separated the real model from 20 independent chemical-block null runs. The resulting prototype is a research decision-support system, not an autonomous assay-termination system. The source is a rat cortical neural MEA assay—not an organ-on-chip dataset—and prospective human neural organ-on-chip validation remains required.

1. Problem and importance

Developmental-neurotoxicity studies follow the maturation of neural cultures over multiple days. Waiting until DIV12 provides the endpoint but delays triage. The useful question is therefore not merely whether DIV12 can be fitted retrospectively, but whether an unseen chemical can be forecast using only information already available by DIV7, whether the forecast beats strong dose- and time-informed comparators, and whether the system knows when not to predict.

MEACompass is a research prototype for that question. It forecasts five functional endpoints as percent of same-plate, same-DIV zero-dose control:

- mean firing rate;
- bursts per minute;
- number of active electrodes (nAE);
- number of network spikes (ns.n); and
- coordinated activity (r).

The interface distinguishes OBSERVED early measurements, PREDICTED DIV12 values and intervals, and HYPOTHESIS statements about later or external use. It never makes an autonomous assay-termination decision. A wide interval produces an abstention recommendation and the assay continues.

Developmental neurotoxicity is difficult to study because neural-network formation is dynamic and functional disturbances can emerge over multiple days. The EPA Network Formation Assay is a new approach methodology that uses microelectrode arrays to follow activity in cortical cultures. Earlier reliable forecasts could prioritize review and allocate confirmatory measurements. They do not replace the final readout, toxicological interpretation, or human oversight.

2. Related work

Frank et al. established medium-throughput MEA measurements for evaluating chemical effects on neural network function. Shafer et al. evaluated network formation across 136 chemicals and several developmental timepoints, providing the longitudinal basis for this study. EPA and OECD work on the DNT in vitro testing battery places these assays in a broader new-approach-method framework; no single assay is a complete developmental-neurotoxicity decision system. MEACompass addresses a narrower problem: forecasting a later functional measurement from causally earlier observations while quantifying uncertainty.

Its distinguishing design choices are chemical-disjoint evaluation, strong time-informed comparators, nested group-aware calibration, a registered negative control with an auditable stop, and selective prediction. Boosted trees suit this tabular, small-chemical regime and remain practical across nested group folds.

3. Data and audit

2.1 Source and scope

The data are from the U.S. EPA Network Formation Assay associated with Shafer et al. (2019). This is explicitly a rat cortical neural microelectrode-array assay, not an organ-on-chip dataset. Its relevance to neural organ-on-chip work is methodological: longitudinal electrophysiology and uncertainty-aware early forecasting can be used in both settings. This study does not demonstrate transfer to a human, microfluidic, or organ-on-chip platform.

2.2 Audited inventory

Measure	Audited value
Official test entries	146
Canonical chemicals	136
Longitudinal records	17,224
Plates/batches	99
Plate-well trajectories	4,344
Complete DIV5/7/9/12 trajectories	4,192 (96.50%)
NTP records	5,712
ToxCast records	11,512
Electrophysiology columns available	18
SMILES resolved	130/136 (95.59%)

The stable experimental key is (cohort, Plate.SN, well). Treatment aliases and biological replicates are canonicalized to CAS RN before splitting. Missing burst-conditional measurements are retained with explicit missingness indicators. The full lineage, checksums, exclusions, and limitations are recorded in docs/meacompass_data_audit.md.

2.3 Sources and licenses

The EPA catalog provides public access and links the EPA ScienceHub terms for the original NFA release. A U.S. Government-work rationale under 17 U.S.C. §105 is applied only when file-level EPA authorship is verified; repository ownership is not treated as proof for every file. The source carries no warranty, and EPA names, seals, or logos must not imply endorsement. Project code is Apache-2.0. Chemical structures were resolved with the public PubChem PUG REST service. Raw EPA data, external-refinement files, caches, and checkpoints are excluded from Git.

4. Preregistered protocol

Primary endpoints, target construction, causal feature contract, group splits, metrics, chemical bootstrap, baseline grid, interval rules, and gates were frozen at commit 8671fd84cc48d5b4fc45fe09ee20227cb3d46ac0 before predictive results. Later decisions are append-only in docs/decisions.md; they do not overwrite the original protocol.

4.1 Time causality

Primary inputs contain raw DIV5 and DIV7 measurements and missingness flags, exposure dose, cohort, and structure-only chemical descriptors. DIV9 is used only in a declared post-lock time ablation. DIV12 measurements, DIV12 controls as features, official potency values, hit calls, and late viability measurements are forbidden. Automated tests fail closed on future-derived feature names.

4.2 Chemical-disjoint nested evaluation

The outer design uses five stratified group folds repeated for seeds 0, 1, and 2. Canonical CAS RN is the group, so every alias, concentration, plate, well, and replicate for a chemical stays on one side of a split. Three-fold group splits inside each outer training set choose baselines and hyperparameters. Outer-test folds are evaluation only. A random well split would leak correlated views of the same chemical and answer an easier, operationally irrelevant question.

4.3 Comparators

- B0: dose-bin mean learned from the outer training data.
- B1/B1b/B2: preregistered temporal and simple reference baselines.
- BT: best of the original preregistered baseline set.
- BT+: best of B0/B1/B1b/B2, selected only on inner validation folds. It was registered after B0 exposed weakness in the original comparator set.
- B3: a simpler gradient-boosted model using early measurements.
- M1: frozen gradient-boosted residual model using DIV5, DIV7, dose, cohort, RDKit descriptors, and Morgan fingerprints.
- BT++: a stricter post-audit comparator selecting between BT+ and a fixed dose-smooth baseline on inner folds only. It was required by the integrity audit and is reported separately from the preregistered comparisons.

4.4 Metrics and uncertainty

The protocol reports DIV12 MAE, RMSE, and Spearman correlation, change-from-DIV7 metrics, paired MAE difference, and relative MAE gain. Confidence intervals use 1,000 resamples of canonical chemical groups. Wells are never bootstrapped as independent observations. Nested group CV+ constructs nominal 90% intervals from training-fold residuals without using the outer test labels for calibration.

5. Deviations and integrity audit

BT+ was added as a registered deviation after the preregistered baseline set was found to omit dose-bin mean B0. BT++ was added only after the bounded integrity audit as a stricter comparison against a fixed dose-smooth model. Both are selected without outer-test labels. Neither change alters endpoints, splits, targets, M1 predictions, or the historical Gate S decision.

The registered chemical-group permutation control unexpectedly showed a small but significant benefit for bursts versus BT+ (paired ΔMAE -0.525 ; 95% CI -0.969 to -0.136). Under the registered rule, Gate S was STOP: suspected leakage or confounding. That decision remains in `results/gate_s_decision.json` and was not rewritten.

A bounded, post-registration audit then tested feature lineage, prediction-target alignment, a full target shuffle, a dose-smooth explanation, and 20 independent chemical-block null runs for B3 and M1 over three seeds and five folds. It found:

- no future-derived or target-derived feature path;
- a clean full-shuffle negative control;
- no prediction/target re-alignment defect;
- generic dose and early-trajectory structure in the original block permutation;
- real M1 performance better than the dose-smooth comparator on all five endpoints, with chemical-bootstrap intervals excluding zero; and
- real M1 performance exceeding every one of the 20 M1 block-null runs on every endpoint (empirical one-sided $p < 0.05$ at the available resolution).

The audit conclusion was `PASS_RESIDUAL_STRUCTURE_UNDER_NULL`, conditional on use of BT++ and retention of the original failed Gate S record. This is a registered audit resolution, not evidence that the initial control was irrelevant.

6. Methods

Each plate-well trajectory is joined across DIV5, DIV7, and DIV12, then normalized to its same-plate, same-DIV zero-dose control. M1 uses DIV5 and DIV7 neural measurements, missingness flags, log-transformed dose, cohort, 16 RDKit physicochemical descriptors, a 2,048-bit Morgan fingerprint, and a structure-missing indicator. Depending on inner validation, a boosted tree either predicts DIV12 directly or predicts the residual around a temporal baseline.

The registered grid varies tree depth, learning rate, minimum child weight, and feature subsampling. Early stopping operates only on inner validation data. Missing measurements are explicit; there is no globally learned imputer or scaler. Confidence intervals and paired comparisons use chemicals, never wells, as the independent resampling unit.

7. Implementation

The implementation uses Python 3.11 with pinned scientific dependencies. Every training task writes a prediction checkpoint and tuning record. Fail-closed tests check time causality, chemical-group disjointness, prediction schemas, identity, terminology, claims, and result-only reproduction. `make reproduce-lite` reads saved evidence only and cannot import a training entry point.

The demonstration consumes the versioned prediction schema and precomputed held-out predictions. It separates early observations, a prediction and interval, and research hypotheses. Its reliability badge renders the frozen abstention rule; it is not a laboratory instruction.

8. Main results

All values below aggregate the repeated chemical-disjoint outer-test predictions. Negative Δ MAE means M1 is better. Confidence intervals are chemical-level paired bootstrap intervals.

Endpoint	M1 MAE	BT+ MAE	Gain vs BT+	Δ MAE vs BT+ (95% CI)	BT++ MAE	Gain vs BT++	Δ MAE vs BT++ (95% CI)
Bursts/min	27.17	32.74	17.0%	-5.57 [-7.51, -3.77]	32.42	16.2%	-5.25 [-7.26, -3.41]
Mean firing rate	39.29	45.79	14.2%	-6.50 [-11.59, -1.50]	44.79	12.3%	-5.49 [-10.48, -0.21]
Active electrodes	20.42	33.50	39.0%	-13.08 [-17.47, -9.52]	35.75	42.9%	-15.33 [-25.82, -7.75]
Network spikes	40.72	48.26	15.6%	-7.54 [-10.20, -4.56]	47.47	14.2%	-6.75 [-9.48, -3.76]
Coordinated activity r	48.61	62.91	22.7%	-14.30 [-17.56, -11.05]	66.03	26.4%	-17.41 [-24.51, -12.32]

M1 significantly beats both BT+ and BT++ on all five endpoints. This is the fair primary claim. Relative gains versus the weaker original BT were 18.3-40.6% and remain available in `results/gate_s_main.csv`, but are not used as the headline.

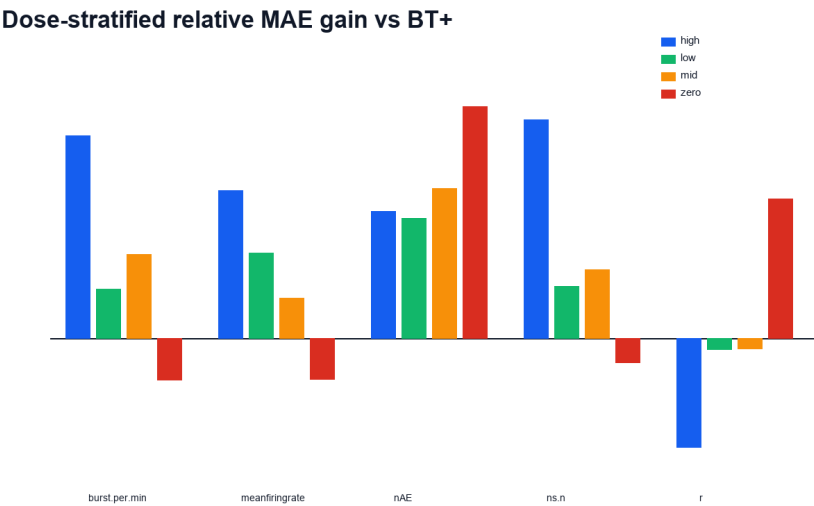
M1 also improved over B3 significantly for active electrodes and coordinated activity; the M1-B3 intervals crossed zero for the other three endpoints. Thus the chemistry-augmented model is not claimed to dominate the simpler model everywhere.

8.1 Dose and cohort robustness

Dose cut points were learned on each outer training fold and applied unchanged to its test fold. At low and mid dose, all five endpoints had beneficial point estimates versus BT+. The confidence intervals were more selective: bursts, active electrodes, and coordinated activity excluded zero at low dose; bursts, active electrodes, network spikes, and coordinated activity excluded zero at mid dose. Firing-rate intervals crossed zero at both low and mid dose, and the network-spike interval crossed zero at low dose. The result is therefore not high-dose-only, but neither is every low/mid subgroup conclusive.

At zero dose, M1 was worse than BT+ by point estimate for bursts (-11.5% relative gain), firing rate (-23.7%), and network spikes (-8.2%). The bursts and network-spike paired intervals excluded zero in the harmful direction; the firing interval crossed zero. Active electrodes and coordinated activity remained beneficial. Interpretation: because the target is percent of matched zero-dose control, simple control-centered baselines may be especially competitive in zero-dose wells. This interpretation is not an established causal explanation. The intended early-triage use is exposed wells, while all zero-dose observations remain visible in the evaluation.

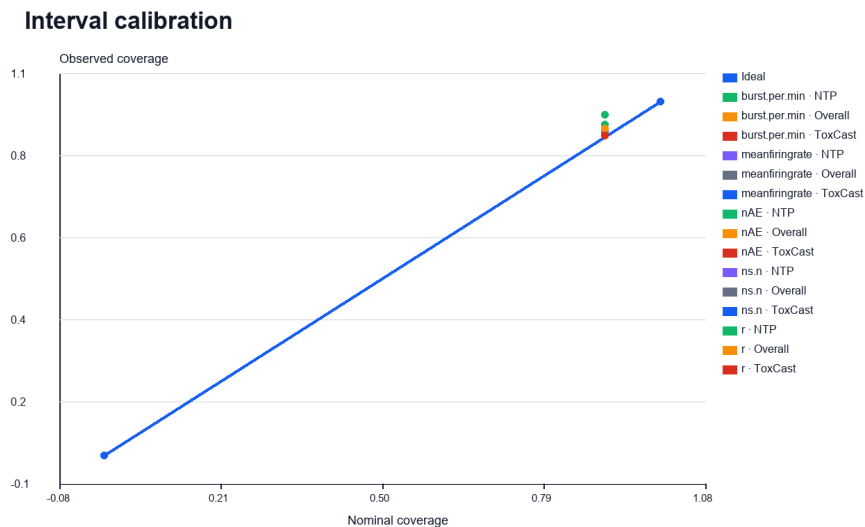
In descriptive cohort slices, M1 had beneficial point estimates versus BT+ for all five endpoints in both NTP and ToxCast. Confidence intervals excluded zero for all five NTP endpoints and four of five ToxCast endpoints; the ToxCast firing-rate interval crossed zero despite a 13.8% point-estimate gain. These are within-study subgroup results, not held-out cohort transfer and not external laboratory or device validation.



Dose-stratified performance

9. Calibration

Nested group CV+ passed Gate M2 without switching to conformalized quantile regression. Empirical coverage for nominal 90% intervals was 91.03–92.40% overall; the lowest separate NTP/ToxCast coverage was 90.36%.



Calibration by endpoint

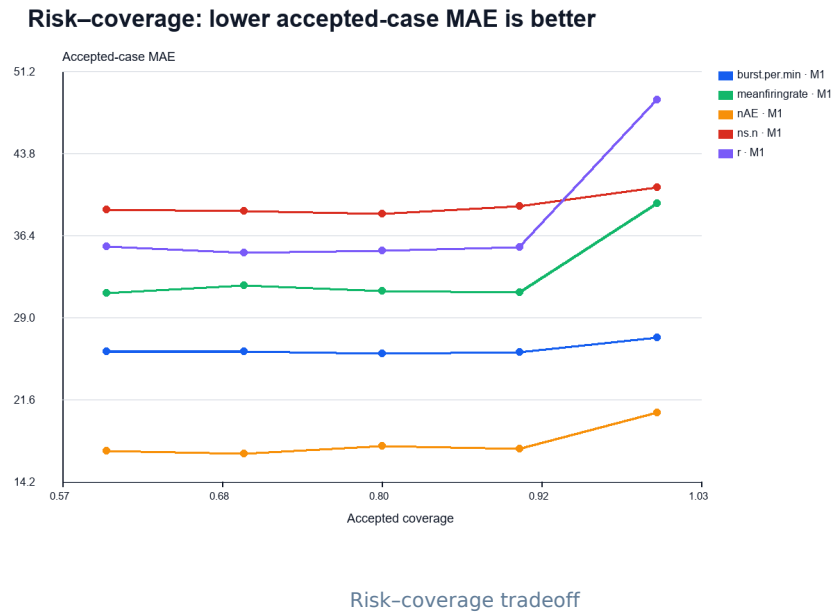
Coverage is reported with interval width because a trivially wide interval is not operationally useful. These retrospective intervals require local recalibration under a platform or population shift.

10. Selective prediction / abstention

At 70% retained coverage, the registered M3 requirement was at least 15% risk reduction with a chemical-bootstrap confidence interval below zero:

Endpoint	Full MAE	Accepted MAE	Risk reduction	Gate M3
Bursts/min	27.17	25.92	4.6%	Limitation: below 15%
Mean firing rate	39.29	31.87	18.9%	Pass
Active electrodes	20.42	16.75	18.0%	Pass
Network spikes	40.72	38.58	5.3%	Limitation: CI crosses zero
Coordinated activity r	48.61	34.81	28.4%	Pass, normalization caveat

M3 therefore passed on three of five endpoints. Bursts and network spikes are reported as limitations rather than omitted. The r result is also qualified: percent-control normalization has an extreme tail when the plate-control denominator is near zero.

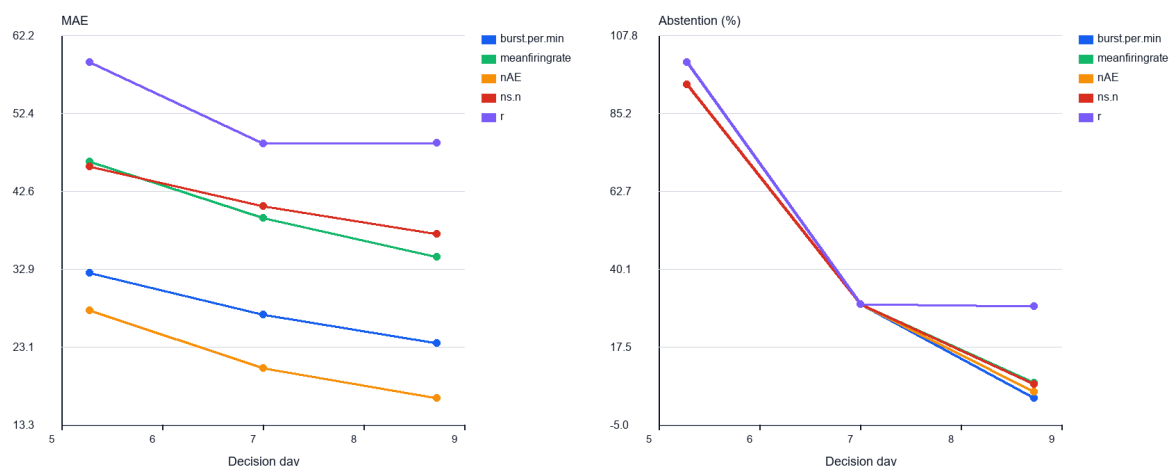


11. Time ablation

The frozen DIV5-only model is substantially weaker than DIV5+DIV7. DIV9 was run only after L1 lock and is an ablation, not a primary result.

Endpoint	DIV5 MAE	DIV5+7 MAE	DIV5+7 gain vs BT+	DIV5+7+9 MAE
Bursts/min	32.39	27.17	17.0%	23.58
Mean firing rate	46.36	39.29	14.2%	34.41
Active electrodes	27.66	20.42	39.0%	16.67
Network spikes	45.78	40.72	15.6%	37.28
Coordinated activity r	58.83	48.61	22.7%	48.74

Accuracy and abstention by decision day



Time ablation

12. Practical value

DIV7 forecasts are available five days before the DIV12 endpoint for accepted cases. With the declared 70% policy, approximately 70% of retrospective held-out instances receive a forecast rather than an abstention. These are time quantities only—no monetary saving is claimed—and prospective workflow validation is required. A forecast can support prioritization but does not conclude the assay.

13. Interpretability

Exact XGBoost contribution values (`pred_contribs`, a TreeSHAP-style additive decomposition) were computed for the 25 frozen seed-0 outer models. DIV7 neural features accounted for 56.9–77.3% of mean absolute contribution, depending on the endpoint. Combined structure features accounted for 11.2–31.8%; DIV5 contributed 3.7–17.6%. Dose and cohort contributions were generally small. This supports the interpretation that late-early physiology drives the forecast while chemistry adds complementary context; it does not establish causal mechanisms.

Five case studies were selected after lock by a disclosed endpoint-scaled error rule: three low-error examples (Fluorene, Glycerol, Picoxystrobin) and two high-error failures (Tributyltin chloride, Mercuric chloride). They are descriptive and are not used to estimate generalization.

14. External validation and cohort-shift robustness

F3b external refinement scoring was CUT at the preregistered harmonization gate. After the deterministic all-2019 three-seed final models and source hashes were frozen, six allowlisted EPA files (20.4 MiB) were acquired from the pinned commit. Across 12,709 overlap recordings, 15 of 18 neural variables passed the compatibility threshold. The refinement objects omit `cv.time` and `cv.network`, while `r` had Spearman 0.817 after the registered transform, below the 0.95 threshold. Because all frozen endpoint models require every registered input, none of the five endpoints was eligible for scoring. No external outcomes were scored and no external-performance claim is made.

The registered fallback held out each 2019 screening cohort in turn, excluded 11 chemicals shared with the source cohort, and used fixed hyperparameters with no target-cohort tuning. M1 significantly beat source-selected BT++ on all five endpoints in both directions. Gains were 10.3–34.5% for ToxCast→NTP and 15.1–50.3% for NTP→ToxCast; every paired chemical-bootstrap interval was below zero. This supports the narrow conclusion: retained predictive value under a held-out NTP↔ToxCast cohort shift without model retuning. It is not external laboratory/device transfer and does not replace the locked headline.

15. Adoption path for neural organ-on-chip research

MEACompass provides an evaluation and adaptation workflow that can be retrained, recalibrated, and prospectively validated on a laboratory's own longitudinal MEA dataset. It is not evidence that the frozen EPA model can be deployed directly on a new chip.

A responsible adaptation maps local electrodes, activity features, timepoints, controls, and missingness to an explicit schema. Control normalization is checked for stability. Chemical- or intervention-disjoint splits reflect intended use, with all replicates grouped. The model is trained locally, intervals are calibrated only on local development data, and coverage is audited across batches, devices, cell donors, and exposure ranges. Drift triggers recalibration or abstention. A prospective study with predeclared criteria and human review is required before operational use.

16. Limitations

- The study is retrospective and covers 125–136 chemical groups by endpoint.
- Rat cortical culture biology cannot establish human neural organ-on-chip transfer.
- NTP/ToxCast subgroup results are descriptive because cohorts were not external held-out domains.
- r percent-control normalization can be unstable near zero denominators.
- Abstention did not reach the registered 15% improvement for bursts or network spikes.
- At zero dose, M1 was worse than BT+ by point estimate for bursts, firing rate, and network spikes. A control-normalization explanation is interpretation, not an established finding; the intended early-triage use is exposed wells.
- Chemistry contributions are predictive associations, not mechanistic evidence; M1 did not significantly beat B3 on three endpoints.
- The original Gate S failed. The later audit narrows the suspected mechanism and supports continuation, but does not erase the registered stop.
- BT+ and BT++ are transparent post-registration safeguards, not preregistered headline comparators.
- A historical B3 tuning-log gap is documented. Held-out predictions were not rerun because no validity defect was found.
- The prototype has not been prospectively validated, monitored under deployment drift, or approved for laboratory decisions.

17. Ethics and compliance

MEACompass is a research decision-support system, not an autonomous assay-termination system. A low-confidence case continues to DIV12; a high-confidence forecast remains subject to expert review. The interface labels observations, predictions, and hypotheses separately to limit automation bias. No human participant data are used. Raw source data and derived per-well external files are not redistributed by this repository.

Claims are bounded to rat cortical MEA data. Human relevance, regulatory acceptance, direct neural organ-on-chip performance, and causal mechanism are outside the evidence. The project does not imply EPA endorsement. Prospective validation, governance, drift monitoring, and a human escalation path are prerequisites for real laboratory use.

18. AI-tool disclosure

Tool	Role	Human verification	Scientific-number source
USER CONFIRMATION REQUIRED	USER CONFIRMATION REQUIRED	Repository tests, source review, and final author review	Executed CSV/JSON result artifacts only

No AI assistant or service name is inferred. The final disclosure will list only tools and roles confirmed directly by the team. Human authors remain responsible for study design, code review, source checking, interpretation, and claims.

19. Reproduction

The result-only reproduction path is:

```
make setup
make test
make reproduce-lite
make demo
```

`reproduce-lite` reads saved prediction/result files only and never imports a training entry point. It fails with an explicit artifact list when required files are missing. The demo loads precomputed held-out predictions and operates offline after dependencies are installed. The frozen prediction contract is `schemas/prediction_schema_v1.yaml`.

Key evidence artifacts are:

- `results/gate_s_main.csv` and `results/bt_plus_plus.csv` — fair baseline results;
- `results/audit/` — bounded integrity audit and null controls;
- `results/m2_calibration.csv` — interval calibration;
- `results/m3_risk_coverage.csv` — selective prediction;
- `results/time_ablation.csv` and `results/practical_value.csv` — time analysis;
- `results/potency_summary.csv` — explicitly non-primary potency appendix; and
- `results/feature_importance.csv` and `results/case_studies.csv` — interpretation.

The environment is declared in `environment.yml`; package dependencies are pinned in `pyproject.toml`. Data acquisition is manifest-driven and verifies hashes. A clean-clone reproduction has passed tests and the result-only figure pipeline. Full training remains separate so judges can verify tables and figures without automatically retraining models.

20. Sources and licenses

Project code is Apache-2.0. The original EPA release is referenced by DOI and its catalog/ScienceHub terms; no raw EPA file is committed. PubChem provides structure lookups and RDKit computes local deterministic descriptors. The F3b refinement repository has no detected root license, so only the amended file-level provenance determination is recorded; no refinement file is redistributed. The file-oriented record is `docs/sources_and_licenses.md`.

21. Conclusion

MEACompass demonstrates a credible retrospective result: DIV7 information can forecast DIV12 neural MEA endpoints for unseen chemicals more accurately than strong inner-selected baselines, with nominal 90% intervals that attain coverage and an abstention mechanism that reduces error materially on three endpoints. Its strongest contribution is the combination of chemical-disjoint evaluation, explicit negative-control failure and audit, calibrated uncertainty, and an honest continuation path for uncertain cases. The correct next scientific step is a frozen, prospective external validation on human neural organ-on-chip data—not a claim that the current rat assay model is ready to control a laboratory.

22. References

- EPA dataset and DOI: <https://doi.org/10.23719/1503191>
- Data.gov catalog: <https://catalog.data.gov/dataset/data-for-evaluation-of-chemical-effects-on-network-formation-in-cortical-neurons-grown-on>
- Shafer et al. (2019): <https://doi.org/10.1093/toxsci/kfz052>
- Frank et al. (2017): <https://doi.org/10.1016/j.tiv.2017.09.016>
- EPA DNT NAM fact sheet: https://www.epa.gov/system/files/documents/2022-10/DNT_Factsheet_10_28_22_Final_1.pdf
- OECD DNT in vitro battery recommendations: <https://doi.org/10.1787/91964ef3-en>
- EPA ScienceHub license: <https://pasteur.epa.gov/license/sciencehub-license.html>
- PubChem PUG REST: <https://pubchem.ncbi.nlm.nih.gov/docs/pug-rest>
- Project license: LICENSE (Apache-2.0)

23. Appendix A — Endpoint and evidence map

Endpoint	Gain vs BT+	Gain vs BT++	M2 coverage	M3 at 70%
Bursts/min	17.0%	16.2%	91.08%	Limitation: 4.6% reduction
Mean firing rate	14.2%	12.3%	91.53%	Pass: 18.9% reduction
Active electrodes	39.0%	42.9%	92.13%	Pass: 18.0% reduction
Network spikes	15.6%	14.2%	91.03%	Limitation: 5.3%, CI crosses zero
Coordinated activity r	22.7%	26.4%	92.40%	Pass: 28.4%, normalization caveat

Primary rows trace to `results/main_results.csv`, BT++ rows to `results/bt_plus_plus.csv`, calibration to `results/m2_calibration.csv`, and risk-coverage to `results/m3_risk_coverage.csv`. F3 secondary results trace to `results/f3_cross_cohort/summary.csv`.

Appendix A.1 Potency target mismatch

Official EPA EC50 values summarize ontogeny area under the curve, whereas this project predicts DIV12 outcomes. Missing official EC50 values are not zeros. An exploratory Hill-grid analysis found rank correlations of 0.773–0.868 between predicted DIV12 curve estimates and available official AUC EC50 values across 54–61 chemicals. Because the targets differ, the result is appendix-only and the project does not claim to predict official potency five days earlier.

24. Appendix B — Registered status ledger

Gate	Status	Meaning
Gate S	STOP, preserved	Initial block permutation showed a small bursts benefit
Bounded integrity audit	PASS	No leakage; full shuffle clean; separation from 20 null runs
M1	PASS	Five-endpoint locked point-prediction result
M2	PASS	Nominal 90% coverage attained for every endpoint
M3	PASS, 3/5	Passes for firing rate, active electrodes, and r
L1	LOCK	Primary model and reliability protocol frozen
F3b	CUT	Required inputs absent / not comparable; no external outcomes scored
F3	PASS, secondary	Five wins in both chemically disjoint cohort directions

This ledger preserves failures, deviations, and boundaries alongside positive results. No appendix result supersedes the locked primary evaluation.